

UNIVERSIDADE FEDERAL DE SANTA CATARINA
CAMPUS FLORIANÓPOLIS
DEPARTAMENTO DE INFORMÁTICA E ESTATÍSTICA
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIA DA COMPUTAÇÃO

Guilherme da Silva Pierri
Jonatan Felipe Hartmann
Rafael dos Santos Oliveira

Traffic Shaping in Zonal In-Vehicle Networks with Time-Sensitive Networking: Project Design

Florianópolis
2026

LIST OF ACRONYMS AND NOTATION

ADAS – Advanced Driver Assistance Systems.

BE – Best Effort.

CBS – Credit-Based Shaper.

CSW – Central TSN Switch.

E/E – Electrical and Electronic.

GCL – Gate Control List.

gPTP – generalized Precision Time Protocol.

HPC – High-Performance Computer.

IEEE – Institute of Electrical and Electronics Engineers.

INET – Communication-network simulation framework for OMNeT++.

IVN – In-Vehicle Network.

OMNeT++ – Objective Modular Network Testbed in C++.

PCP – Priority Code Point.

SP – Strict Priority.

SR-A/SR-B – Stream Reservation classes A and B.

TAS – Time-Aware Shaper.

TC – Traffic Class.

TSN – Time-Sensitive Networking.

UDP – User Datagram Protocol.

VLAN – Virtual Local Area Network.

Z1–Z3 – Physical zones 1 through 3.

ZCPU_i – Zonal processor in zone i.

ZSW_i – Zonal switch in zone i.

CONTENTS

1 INTRODUCTION....4
1.1 CONTEXT AND MOTIVATION....4
1.2 PROBLEM STATEMENT AND RESEARCH GAP....4
1.3 PROJECT GOAL AND SPECIFIC OBJECTIVES....5
1.4 RESEARCH QUESTIONS AND HYPOTHESES....5
1.5 PROJECT SCOPE....6
2 SYSTEM ARCHITECTURE....6
2.1 ARCHITECTURE BASIS AND DESIGN RATIONALE....6
2.2 LOGICAL ARCHITECTURE AND COMPONENTS....7
2.3 INET MODEL MAPPING....8
2.4 LINKS PATHS AND BOTTLENECKS....9
2.5 QUEUEING SYNCHRONIZATION AND FORWARDING MODEL....10
2.6 MODELING ASSUMPTIONS....11
3 TRAFFIC MODEL....12
3.1 TRAFFIC MODELING CONVENTIONS....12
3.2 TRAFFIC FLOW CATALOG....13
3.3 PERIODIC BURSTY AND TRANSIENT TRAFFIC....16
3.4 LOAD LEVELS AND TEMPORAL AGGREGATION....18
3.5 PER LINK LOAD ANALYSIS....19
3.6 TRAFFIC MODEL DECISIONS....20
4 TSN CONFIGURATION DESIGN....21
4.1 COMMON CLASSIFICATION AND QUEUEING RULES....21
4.2 STRICT PRIORITY CONFIGURATIONS....21
4.3 CREDIT-BASED SHAPING CONFIGURATIONS....22
4.4 CBS CALIBRATION....22
4.5 TIME-AWARE SHAPING WITH CREDIT-BASED SHAPING....23
4.6 CONFIGURATION COMPARISON....23
5 EXPERIMENTAL DESIGN....24
5.1 EXPERIMENTAL FACTORS AND MATRIX....24
5.2 RUN PROTOCOL....25
5.3 CALIBRATION, PILOT, AND FINAL CAMPAIGN....25
5.4 METRICS AND VIABILITY CRITERIA....25
5.5 HYPOTHESIS TESTING AND DATA ANALYSIS....26
6 IMPLEMENTATION AND VALIDATION PLAN....27
6.1 IMPLEMENTATION SCOPE....27
6.2 VERIFICATION AND PILOT....27
6.3 REPRODUCIBILITY AND HANDOFF....28
7 CONCLUSION....28
REFERENCES....29

1 INTRODUCTION

1.1 CONTEXT AND MOTIVATION

Zonal electrical/electronic architectures connect sensors and actuators to nearby zonal switches and concentrate communication toward central computing platforms. This can reduce cabling and support consolidation, but it also creates shared links where control, sensor, multimedia, transient ADAS, and best-effort traffic compete (COLUSSI, 2025; LUO et al., 2022). Because these flows differ in size, period, burstiness, and deadline, average link utilization alone does not describe the resulting queueing behavior.

Time-Sensitive Networking (TSN) provides mechanisms for differentiating and controlling this traffic. Virtual Local Area Network (VLAN) tagging and Priority Code Point (PCP) classification are specified by IEEE 802.1Q; Credit-Based Shaping (CBS) was introduced by IEEE 802.1Qav; Time-Aware Shaping (TAS) by IEEE 802.1Qbv; and synchronized operation is supported by IEEE 802.1AS. IEEE 802.1DG profiles TSN capabilities for automotive in-vehicle Ethernet (IEEE, 2009; IEEE, 2015; IEEE, 2020; IEEE, 2022; IEEE, 2025).

The project compares these mechanisms under a fixed zonal workload. Strict Priority (SP) is simple but may penalize lower classes; source pre-shaping reduces camera microbursts; CBS limits and reserves service by traffic class; and TAS creates synchronized transmission windows. The relevant question is therefore not which mechanism is universally best, but which trade-offs appear as load and burst alignment change.

1.2 PROBLEM STATEMENT AND RESEARCH GAP

Navet et al. (2017) and Migge et al. (2018) compared SP, source pre-shaping, CBS, and TAS/CBS combinations in automotive Ethernet, while Luo et al. (2022) studied TSN in a zonal architecture. Leal Filho et al. (2025) further showed that transient ADAS transfers can create short overloads that are hidden by average-load analysis.

A controlled comparison that combines these traffic treatments with explicit zonal convergence points, several offered-load levels, and aligned versus staggered bursts remains missing. This project addresses that gap with one fixed three-zone benchmark and a common Traffic Catalog.

1.3 PROJECT GOAL AND SPECIFIC OBJECTIVES

The goal is to evaluate how zonal aggregation, offered load, and burst alignment affect five configurations: C0 uses SP; C1 adds camera source pre-shaping; C2 uses nominal CBS; C3 uses calibrated CBS; and C4 combines calibrated CBS with TAS.

The specific objectives are to define the architecture and 22-flow workload; implement comparable configurations in OMNeT++/INET; evaluate four load levels (L0–L3) and two aggregation patterns (A0 staggered and A1 aligned); collect latency, deadline, throughput, loss, utilization, and queue metrics; and relate the results to predefined hypotheses.

1.4 RESEARCH QUESTIONS AND HYPOTHESES

RQ1 – Which conclusions from previous automotive Ethernet studies remain valid in a zonal network?

RQ2 – How does burst alignment affect tail latency, deadline misses, and queue occupancy at equal average load?

RQ3 – When do source pre-shaping and calibrated CBS improve on SP and nominal CBS?

RQ4 – What additional benefit and cost does TAS provide over calibrated CBS?

RQ5 – Does any configuration remain preferable across every load and aggregation regime?

H1 – At low load, differences among C0–C4 will be small for critical traffic.

H2 – Aligned bursts will worsen tail latency and queue occupancy under SP, while source pre-shaping will reduce this effect.

H3 – Within the 75% per-port reservation cap, calibrated CBS will extend the viability frontier beyond nominal CBS without unacceptable loss of best-effort service.

H4 – TAS will improve brake-traffic timing under high load, with costs in usable capacity, synchronization, and configuration effort.

H5 – No single configuration will dominate all metrics in all operating regimes.

1.5 PROJECT SCOPE

The scope comprises one fixed three-zone Ethernet topology, 22 flows, configurations C0–C4, loads L0–L3, patterns A0–A1, and the simulation and analysis required to compare them. The primary outputs are latency and deadline behavior for critical flows, service delivered to best-effort traffic, queue and link behavior, and the main configuration trade-offs.

FRER, path redundancy, failure injection, ATS, PSFP, frame preemption, dynamic routing, automatic topology or GCL synthesis, formal worst-case analysis, and complete vehicle-function modeling are outside the scope. The study provides comparative evidence for this benchmark rather than a conformance assessment or a universal vehicle-network design.

2 SYSTEM ARCHITECTURE

2.1 ARCHITECTURE BASIS AND DESIGN RATIONALE

The project adopts the three-zone benchmark, a tree-topology zonal IVN composed of three physical zones, one central TSN switch, and one central vehicle computer. Its high-level organization is derived from the Tree architecture proposed by Colussi (2025), particularly the separation among the vehicle central system-on-chip, the central switch, the zonal switches, and the zonal processors. The allocation of sensors, actuators, and application families is complemented with the automotive traffic architecture of Luo et al. (2022). Transient ADAS communication is motivated by Leal Filho et al. (2025), while the heterogeneous combination of critical, multimedia, and best-effort traffic follows the experimental concerns raised by Navet et al. (2017) and Migge et al. (2018).

The tree was selected because it provides unique Layer 2 paths and explicit aggregation points. Local traffic first converges at a zonal switch, traffic exchanged with the central computer additionally converges at the CSW, and cross-zone communication traverses two zonal switches and the central backbone. These properties make queue buildup and contention directly observable without introducing routing alternatives that could confound the comparison among traffic-shaping mechanisms.

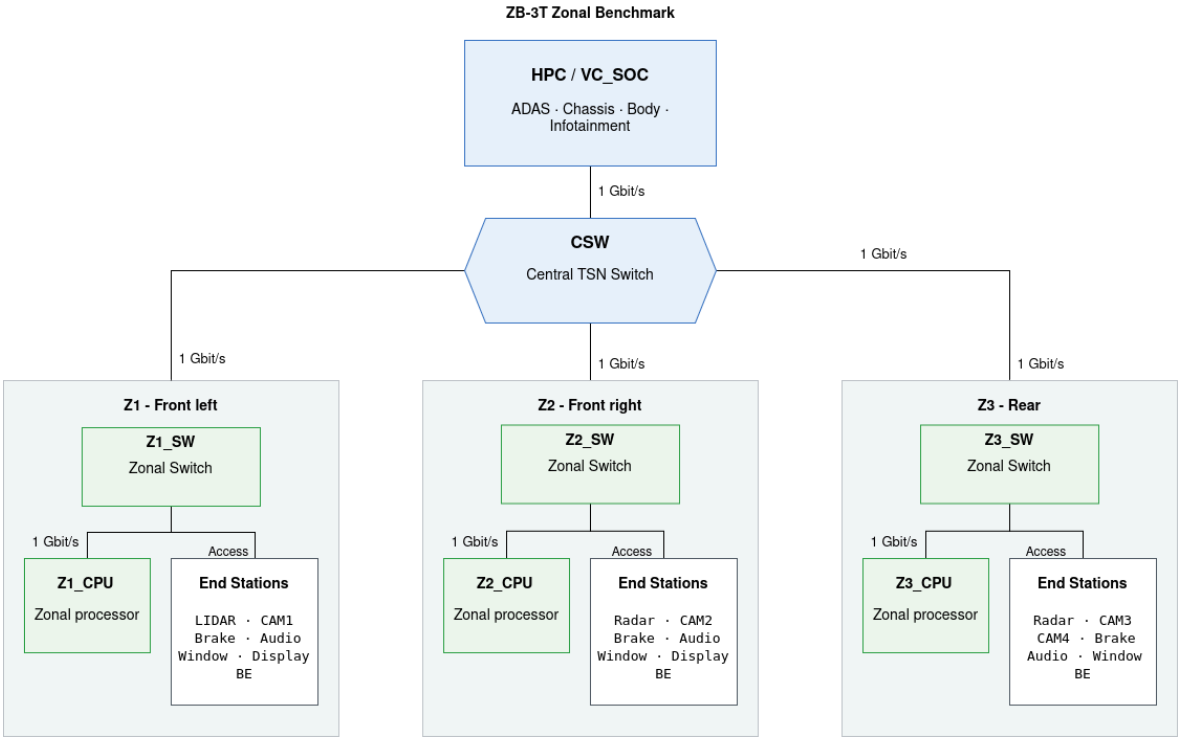
The three zones represent physical locations rather than functional domains. ADAS, chassis, body, infotainment, and diagnostic functions may execute on the HPC or on a zonal processor, but they do not constitute separate domain networks. The topology, routes, link capacities, application endpoints, and traffic catalog remain fixed across C0–C4; only the

declared classification, shaping, and gating configuration changes. Ring topologies, redundant paths, reconvergence, and failure recovery are deliberately excluded from this benchmark.

2.2 LOGICAL ARCHITECTURE AND COMPONENTS

The logical topology is shown in Figure 1. The HPC connects to the CSW, which provides one backbone branch to each zonal switch. Within every zone, the zonal processor and all local end stations connect directly to the respective zonal switch. Consequently, a zonal processor participates in a communication path only when it is the source or destination of that flow; it is not an obligatory forwarding hop for frames exchanged between local end stations and the central backbone.

Figure 1 – Logical architecture of the three-zone benchmark.



Source: the authors, based on Colussi (2025) and Luo et al. (2022).

The benchmark contains the following logical components:

- **HPC / VC_SOC** – Central vehicle computer hosting consolidated ADAS, chassis, body, infotainment, and diagnostic applications.
- **CSW** – Central TSN switch connecting the HPC to the three zonal branches and concentrating backbone traffic.

- **Z1** – Front-left zone composed of Z1_SW, Z1_CPU, LiDAR1, CAM1, BRAKE1, WINDOW1, AUDIO1, DISPLAY1, and BE1.
- **Z2** – Front-right zone composed of Z2_SW, Z2_CPU, RADAR1, CAM2, BRAKE2, WINDOW2, AUDIO2, DISPLAY2, and BE2.
- **Z3** – Rear zone composed of Z3_SW, Z3_CPU, RADAR2, CAM3, CAM4, BRAKE3, WINDOW3, AUDIO3, and BE3.
- **SYNC** – Logical time reference implemented by the HPC clock, which acts as the gPTP grandmaster in C4 and synchronizes the CSW and the three zonal switches.

Sensors, actuators, audio devices, displays, and best-effort nodes are modeled as communication endpoints. Their application payloads and timing behavior are represented, but their internal sensing, control, perception, rendering, and actuation functions are outside the network model.

2.3 INET MODEL MAPPING

The logical architecture is mapped to INET modules so that the model preserves the distinction between forwarding elements, processing endpoints, and application traffic:

- **HPC and ZCPU_i** – Represented by TsnDevice modules containing multiple source and destination applications. These devices terminate their own traffic and do not forward frames addressed to other nodes. In C4, the HPC additionally enables egress traffic shaping on its interface toward the CSW.
- **CSW and ZSW_i** – Represented by TsnSwitch modules. Traffic classification, egress queues, and CBS are configured as required by C0–C4; C4 additionally enables TAS on the three CSW egresses toward ZSW_i and on the three ZSW_i egresses toward BRAKE_i.
- **ZonalController** – A project-specific compound module that groups one TsnSwitch and one TsnDevice connected by the corresponding LP link. This composition represents a zonal switch and processor without inserting the processor into unrelated Ethernet paths.
- **Sensors, actuators, audio nodes, displays, and best-effort nodes** – Represented by TsnDevice modules acting as abstract traffic sources or sinks.
- **Ethernet links** – Represented by parameterized Ethernet channels whose data rate and propagation delay are selected from the link groups defined in Section 2.4.

- **Traffic applications** – Implemented as UDP source and destination applications. Packet size, period or arrival distribution, initial offset, destination, and flow identifier are supplied by the Traffic Catalog in Section 3.
- **Synchronization** – Implemented in C4 with INET clock and gPTP modules. The HPC acts as the time master, while the CSW and ZSW_i participate in the same gPTP domain; the configuration and protocol overhead are fixed across all C4 runs.

This mapping keeps the workload independent from the selected traffic treatment. A scenario changes configuration parameters rather than replacing application or topology modules, which reduces the risk that implementation differences are mistaken for effects of the scheduling mechanism.

2.4 LINKS PATHS AND BOTTLENECKS

Links are grouped by architectural role so that capacity and delay assumptions can be applied consistently:

- **LC** – One full-duplex connection between the HPC and CSW, operating at 1 Gbit/s with 100 ns of propagation delay. This is the principal central convergence link.
- **LB** – Three full-duplex backbone links between the CSW and ZSW_i, each operating at 1 Gbit/s with 100 ns of propagation delay.
- **LP** – Three full-duplex internal links between ZSW_i and ZCPU_i, each operating at 1 Gbit/s with 100 ns of propagation delay.
- **LA-H** – Five full-duplex high-rate access links connecting LiDAR1 and CAM1–CAM4 to their zonal switches, each operating at 1 Gbit/s with 100 ns of propagation delay.
- **LA-L** – Sixteen full-duplex access links connecting the remaining end stations to their zonal switches, each operating at 100 Mbit/s with 100 ns of propagation delay.

The 1 Gbit/s backbone capacity is consistent with the automotive architecture reported by Luo et al. (2022), while the access-link differentiation is a controlled adaptation that prevents unintended bottlenecks at high-rate sensors and retains realistic lower-capacity access for other devices. The uniform 100 ns propagation delay corresponds to approximately 20 m at a propagation speed of 2×10^8 m/s. Because the links are full-duplex, utilization and offered load are evaluated independently in each direction.

Four recurring path patterns are defined:

- **P1** – Sensor-to-central-processing path: sensor → ZSW_i → CSW → HPC.
- **P2** – Central-control and infotainment path: HPC → CSW → ZSW_i → actuator, audio node, or display.
- **P3** – Zonal-processing path: sensor → ZSW_i → ZCPU_i.
- **P4** – Cross-zone ADAS path: ZCPU_i → ZSW_i → CSW → ZSW_j → ZCPU_j.

The design deliberately creates four categories of bottlenecks:

- **G1** – Each zonal uplink, where local sensor and processor traffic converges before entering the backbone.
- **G2** – The CSW-to-HPC direction of LC, where traffic arriving from the three zones converges at the HPC ingress.
- **G3** – Each zonal downlink, where control, audio, video, and best-effort traffic share the path from the CSW to a zone.
- **G4** – The LP and backbone links traversed by transient cross-zone ADAS transfers, particularly the converging paths toward Z3_CPU.

These link groups, paths, and bottlenecks are fixed identifiers used by the Traffic Catalog, configuration rules, instrumentation, and result analysis.

2.5 QUEUEING SYNCHRONIZATION AND FORWARDING MODEL

All switches use store-and-forward operation. No additional fixed processing delay is imposed by the project model, but the serialization, reception, MAC, bridge, queueing, and transmission behavior implemented by INET remains active. Layer 2 forwarding paths are static; STP, dynamic routing, link failures, physical transmission errors, and error-triggered retransmissions are not modeled. Multicast replication, when required by the workload, is performed by the appropriate switch rather than by duplicating traffic at the source.

Frames are classified through VLAN and PCP information and placed in per-class egress queues. The topology and queue capacity are common to C0–C4, while the service rule applied to those queues changes according to the selected configuration. Queues use drop-tail behavior, with 128 KiB per class at the deliberate bottlenecks G1–G4, 64 KiB per class at the camera talker egress ports carrying F02–F05, and 32 KiB per class at the remaining output ports. One configured camera burst contains 30 frames of 1,496 bytes at the MAC level, totaling 44,880 bytes; the 64 KiB source queues can admit the complete burst without source-side overflow. This prevents source buffering from truncating the intended workload

before it reaches the switches; switch buffer overflow remains part of the measured network behavior. The larger value accommodates the derived 90,960-byte upper bound produced by four aligned camera bursts at the CSW-to-HPC output and provides approximately 44% additional margin without making the buffers effectively unbounded.

Configuration C4 uses one in-band gPTP domain to synchronize all independently clocked devices that execute scheduled transmission. The logical SYNC reference is implemented by the HPC clock, configured as the grandmaster; the CSW and Z1_SW–Z3_SW operate as time-aware bridges in the same domain. Automatic grandmaster selection, redundant time domains, and synchronization failures are outside the scope. INET generates periodic two-step Sync/Follow_Up messages and peer-delay exchanges over LC and the three LB links, using fixed synchronization and peer-delay intervals of 125 ms and 1 s, respectively. These control frames share the Ethernet links with application traffic, are assigned to TC7, and consume transmission time and TC7 gate capacity where TAS is active, but are not included among the 22 application flows. The BRAKE1–BRAKE3 end stations are receivers only and therefore do not require synchronized clocks or TAS. C0–C3 do not activate gPTP (IEEE, 2020).

Together, these decisions ensure that measured delay includes transmission, serialization, switching, and queueing effects, while excluding application execution, sensing, physical actuation, and unspecified device-processing delays.

2.6 MODELING ASSUMPTIONS

The topology, Layer 2 paths, link capacities, and endpoint allocation are fixed across C0–C4. Zones represent physical placement, and a zonal processor appears in a path only when it is the source or destination.

The benchmark is Ethernet-only and uses static forwarding. CAN/LIN conversion, routing changes, failures, physical transmission errors, and recovery mechanisms are outside the model. The latency metric covers network traversal and excludes sensing, application execution, and physical actuation.

High-rate sensors use 1 Gbit/s access links and the remaining end stations use 100 Mbit/s links. Switches are store-and-forward, queues are finite and drop-tail, and full-duplex directions are evaluated independently.

F02–F05 use the bursty camera packetization from Migge et al. (2018); F14 is a fixed multicast video stream; F18–F20 are controlled best-effort UDP flows; and F21–F22 preserve the empirical duration–volume pairs from Leal Filho et al. (2025). Reported values are distinguished from derived conversions and benchmark assumptions in the Traffic Catalog.

3 TRAFFIC MODEL

3.1 TRAFFIC MODELING CONVENTIONS

The Traffic Catalog uses a single normalization convention so that source values reported at different protocol layers can be compared and converted into physical link occupancy. The source codes used throughout the catalog are: L6 and L8 for Tables 6 and 8 of Luo et al. (2022); M18-T1 and M18-F6 for Table 1 and Figure 6 of Migge et al. (2018); LF-I and LF-II for Tables I and II of Leal Filho et al. (2025); A for an assumed value; and D for a derived value.

For flows derived from Luo et al. (2022), the reported Size is interpreted as the IPv4 datagram length. The INET UDP application therefore uses S_app = Size − 28 bytes. A VLAN-tagged Ethernet frame has S_MAC = Size + 22 bytes, which reproduces the bandwidth values reported in the original table. Physical occupancy additionally includes 8 bytes of preamble and Start Frame Delimiter and 12 bytes of interframe gap, resulting in S_wire = S_MAC + 20 bytes. For N packets emitted during period T, the normalized link rate is R_wire = N × S_wire × 8 / T. This layer interpretation is derived from numerical consistency with the source and will be verified during implementation.

Migge et al. (2018) explicitly report application payload sizes. For camera traffic, S_app = 1,446 bytes, S_IP = 1,474 bytes, S_MAC = 1,496 bytes, and S_wire = 1,516 bytes; each image contains 30 payloads. For F21–F22, the empirical volume from Leal Filho et al. (2025) is treated as application payload, with 1 MB = 10^6 bytes. The sender segments this payload into UDP payloads of at most 1,472 bytes, producing IPv4 datagrams of at most 1,500 bytes and a maximum physical occupancy of 1,542 bytes per full packet under the same convention.

For F01, F06, and F07, the hard real-time classification is derived from Luo et al. (2022), which assigns their PCP 6 safety-related media class high criticality, no tolerance to loss, and an end-to-end delay limit of 100 µs over five hops. The source does not explicitly

use the hard/soft terminology for these flows; therefore, the hard label is marked as derived (D) in the catalog.

Every catalog value is classified as reported, derived, or assumed. Reported values retain their source identifier; derived values retain the conversion rule and inputs; assumed values are justified by the benchmark design. Rates in this section are wire-occupancy rates and include MAC overhead, preamble/SFD, and IFG.

3.2 TRAFFIC FLOW CATALOG

The Traffic Catalog contains 22 flows. Table 1 fixes each flow family, endpoint pair, path pattern, and architectural provenance. Tables 2a and 2b then provide the complete timing, quality-of-service, classification, nominal-rate, and provenance fields.

Table 1 – Structural Traffic Catalog.

ID	Family	Source	Destination	Path and provenance
F01	LiDAR	LiDAR1	HPC	P1-Z1; endpoint L6; adapted route
F02	ADAS camera	CAM1	HPC	P1-Z1; Luo + Migge workload
F03	ADAS camera	CAM2	HPC	P1-Z2; Luo + Migge workload
F04	Vision camera	CAM3	HPC	P1-Z3; Luo + Migge workload
F05	Vision camera	CAM4	HPC	P1-Z3; Luo + Migge workload
F06	Radar	RADAR1	HPC	P1-Z2; L6; adapted route
F07	Radar	RADAR2	HPC	P1-Z3; L6; adapted route
F08	Brake	HPC	BRAKE1	P2-Z1; L6; consolidated central source
F09	Brake	HPC	BRAKE2	P2-Z2; L6; consolidated central source
F10	Brake	HPC	BRAKE3	P2-Z3; L6; consolidated central source
F11	Audio	HPC	AUDIO1	P2-Z1; L6; consolidated central source
F12	Audio	HPC	AUDIO2	P2-Z2; L6; consolidated central source
F13	Audio	HPC	AUDIO3	P2-Z3; L6; consolidated central source
F14	Multicast video	HPC	DISPLAY1 and DISPLAY2	P2-Z1/P2-Z2; L6 + assumed multicast
F15	Window	HPC	WINDOW1	P2-Z1; L6; consolidated central source

ID	Family	Source	Destination	Path and provenance
F16	Window	HPC	WINDOW2	P2-Z2; L6; consolidated central source
F17	Window	HPC	WINDOW3	P2-Z3; L6; consolidated central source
F18	Best effort	HPC	BE1	P2-Z1; N4 family; assumed instance
F19	Best effort	HPC	BE2	P2-Z2; N4 family; assumed instance
F20	Best effort	HPC	BE3	P2-Z3; N4 family; assumed instance
F21	Transient ADAS	Z1_CPU	Z3_CPU	P4 Z1→Z3; LF-II; assumed convergence
F22	Transient ADAS	Z2_CPU	Z3_CPU	P4 Z2→Z3; LF-II; assumed convergence

Source: the authors, based on Luo et al. (2022), Migge et al. (2018), and Leal Filho et al. (2025).

Table 2a – Traffic timing and quality-of-service parameters.

ID	Normalized size	Emission and offset	Deadline	PCP / criticality
F01	1,500 B IP; 1,542 B wire	1/100 µs; off 0	100 µs / 5 hops	6 / high-hard (D)
F02	30×1,446 B app; 1,516 B wire	burst / 16 ms; A0/A1	10 ms	4 / high-hard
F03	30×1,446 B app; 1,516 B wire	burst / 16 ms; A0/A1	10 ms	4 / high-hard
F04	30×1,446 B app; 1,516 B wire	burst / 16 ms; A0/A1	30 ms	4 / medium-soft
F05	30×1,446 B app; 1,516 B wire	burst / 16 ms; A0/A1	30 ms	4 / medium-soft
F06	42 B IP; 84 B wire	1/1 ms; off 20 µs	100 µs / 5 hops	6 / high-hard (D)
F07	42 B IP; 84 B wire	1/1 ms; off 25 µs	100 µs / 5 hops	6 / high-hard (D)
F08	42 B IP; 84 B wire	1/100 µs; off 0	100 µs / 5 hops	7 / high-hard
F09	42 B IP; 84 B wire	1/100 µs; off 2 µs	100 µs / 5 hops	7 / high-hard
F10	42 B IP; 84 B wire	1/100 µs; off 4 µs	100 µs / 5 hops	7 / high-hard
F11	234 B IP; 276 B wire	1/200 µs; off 0	10 ms soft	4 / medium-soft
F12	234 B IP; 276 B wire	1/200 µs; off 5 µs	10 ms soft	4 / medium-soft
F13	234 B IP; 276 B wire	1/200 µs; off 10 µs	10 ms soft	4 / medium-soft
F14	1,500 B IP; 1,542 B wire	1/200 µs; off 0	30 ms soft	4 / medium-soft
F15	42 B IP; 84 B wire	1/200 µs; off 50 µs	50 ms	3 / medium
F16	42 B IP; 84 B wire	1/200 µs; off 55 µs	50 ms	3 / medium
F17	42 B IP; 84 B wire	1/200 µs; off 60 µs	50 ms	3 / medium

ID	Normalized size	Emission and offset	Deadline	PCP / criticality
F18	1,500 B IP; 1,542 B wire	1/616.8 µs; phase 0	goodput ratio ≥ 0.95	0 / low
F19	1,500 B IP; 1,542 B wire	1/616.8 µs; phase T/3	goodput ratio ≥ 0.95	0 / low
F20	1,500 B IP; 1,542 B wire	1/616.8 µs; phase 2T/3	goodput ratio ≥ 0.95	0 / low
F21	≤1,472 B app; ≤1,500 B IP; ≤1,542 B wire	event; LF-II pairs; A0/A1	10 ms soft (A)	4 / high (A)
F22	≤1,472 B app; ≤1,500 B IP; ≤1,542 B wire	event; LF-II pairs; A0/A1	10 ms soft (A)	4 / high (A)

Source: the authors, based on Luo et al. (2022), Migge et al. (2018), and Leal Filho et al. (2025).

Table 2b – VLAN allocation, nominal wire rate, and parameter provenance.

ID	VLAN	R_wire (Mbit/s)	Source / status
F01	1	123.360	L6, L8; D
F02	2	22.740	M18-T1; VLAN L6; PCP adapted; A,D
F03	7	22.740	M18-T1; VLAN L6; PCP adapted; A,D
F04	13	22.740	M18-T1; VLAN L6; PCP adapted; A,D
F05	14	22.740	M18-T1; VLAN L6; PCP adapted; A,D
F06	6	0.672	L6, L8; D
F07	12	0.672	L6, L8; D
F08	3	6.720	L6, L8; D
F09	8	6.720	L6, L8; D
F10	15	6.720	L6, L8; D
F11	4	11.040	L6; deadline M18-T1; A,D
F12	9	11.040	L6; deadline M18-T1; A,D
F13	16	11.040	L6; deadline M18-T1; A,D
F14	10	61.680	L6; deadline M18-T1; multicast A,D
F15	5	3.360	L6/L8; lower period A
F16	11	3.360	L6/L8; lower period A
F17	17	3.360	L6/L8; lower period A
F18	18	20.000	M18-T1 (≤20); UDP A,D
F19	19	20.000	M18-T1 (≤20); UDP A,D
F20	20	20.000	M18-T1 (≤20); UDP A,D
F21	21	101.786 active; max	LF-II; PCP adapted; D=10 ms by analogy M18; A,D

ID	VLAN	R_wire (Mbit/s)	Source / status
		111.99	
F22	22	101.786 active; max 111.99	LF-II; PCP adapted; D=10 ms by analogy M18; A,D

Source: the authors, based on Luo et al. (2022), Migge et al. (2018), and Leal Filho et al. (2025).

Together, the three tables form the authoritative flow specification used by application generation, forwarding configuration, instrumentation, load calculations, and the experimental scenarios. F21–F22 are event-driven disturbances; their listed nominal rate applies only while an empirical signature is active.

3.3 PERIODIC BURSTY AND TRANSIENT TRAFFIC

The workload combines periodic, bursty, multicast, best-effort, and transient communication without changing endpoints or packetization among C0–C4. Periodic flows include LiDAR, radar, brake, audio, window, and baseline best-effort traffic. Their periods and initial offsets are deterministic within a scenario; offsets prevent unintended simultaneous release except where A1 explicitly requires alignment.

Camera flows F02–F05 are bursty. Each camera releases 30 application payloads of 1,446 bytes per image with a base cycle of 16 ms. F02–F03 represent ADAS cameras with a 10 ms hard deadline, while F04–F05 represent vision cameras with a 30 ms soft deadline. The burst structure is preserved in C0 and C2–C4; only C1 introduces the source pre-shaping defined in Section 4.

F14 is a multicast video flow from the HPC to DISPLAY1 and DISPLAY2. It crosses LC once and is replicated at the CSW toward LB1 and LB2. Its nominal rate remains fixed at 61.680 Mbit/s in every load level so that each 100 Mbit/s display access link remains feasible.

F18–F20 are controlled unidirectional UDP best-effort flows, one per zone, each offering 20 Mbit/s in normalized wire rate at L1. ACKs, congestion control, and TFTP dynamics are excluded so that the measured service reflects queueing and shaping rather than transport adaptation. Operational viability is load-relative: for each flow, the ratio of received to offered application-payload bytes during measurement must be at least 0.95, with no more than 1% packet loss. At L1, the 0.95 threshold corresponds to 19 Mbit/s on the same normalized-rate basis; no fixed 19 Mbit/s threshold is applied at L0, L2, or L3.

F21 and F22 model transient ADAS transfers from Z1_CPU and Z2_CPU to Z3_CPU. Each occurrence samples with replacement one of the ten empirical duration–volume pairs in Table 3 and uses the measured duration rather than a predicted value. During every measured second, F21 starts 5 ms after the beginning of the block. Under A0, F22 starts 10 ms after F21 ends; under A1, both events start together. For a given seed, the sampled sequence is identical across C0–C4. The active rates are computed as R_app = 8 × V_app / duration and, for the table values, R_wire ≈ R_app × 1,542 / 1,472. The implementation segments each transfer into UDP payloads of at most 1,472 bytes and accounts for the final, possibly shorter datagram exactly.

Table 3 – Empirical transient ADAS signatures used by F21–F22.

Pair	Transition	Measured duration (ms)	V_app (MB)	R_app (Mbit/s)	R_wire (Mbit/s)
LF1	Left→Front (source trace)	59.263	0.712	96.114	100.685
LF2	Left→Front (source trace)	486.459	6.129	100.794	105.587
LF3	Left→Front (source trace)	424.599	5.499	103.608	108.535
LF4	Left→Front (source trace)	145.553	1.945	106.903	111.986
LF5	Left→Front (source trace)	131.821	1.759	106.751	111.827
FR1	Front→Right (source trace)	132.532	1.474	88.975	93.206
FR2	Front→Right (source trace)	186.909	2.133	91.296	95.637
FR3	Front→Right (source trace)	286.140	3.286	91.871	96.240
FR4	Front→Right (source trace)	244.375	2.803	91.761	96.124
FR5	Front→Right (source trace)	181.053	2.118	93.586	98.036

Source: the authors, calculated from Table II of Leal Filho et al. (2025).

The Left→Front and Front→Right labels identify the empirical trace families reported by Leal Filho et al. (2025); they do not define the benchmark network routes. In the zonal model, F21 and F22 are adapted as Z1_CPU→Z3_CPU and Z2_CPU→Z3_CPU, respectively, to create controlled convergence toward the rear zone.

The permanent-flow rates in Tables 2a–2b represent steady operation. For F21–F22, the nominal value represents only the interval in which the selected signature is active; their offered load is zero outside that interval.

3.4 LOAD LEVELS AND TEMPORAL AGGREGATION

The experimental load factor changes packet periods or camera cycles without changing packet size, route, priority, VLAN, or deadline. Table 4 defines the four offered-load levels and the two temporal aggregation patterns.

Table 4 – Offered-load levels and temporal aggregation patterns.

Factor	Value	Operational rule	Provenance / rationale
L0	k = 0.5	$T' = T/k$ for scalable flows	A; low load for H1
L1	k = 1.0	Nominal catalog workload	Normalized references
L2	k = 2.5	$T' = T/2.5$ for scalable flows	A; intermediate regime
L3	k = 4.0	$T' = T/4$ for scalable flows	A; LC ingress ≈ 86.3%; CBS cap boundary
A0	Staggered	Cameras at 0, C/4, C/2, 3C/4; events do not overlap	A; aggregation control
A1	Aligned	Cameras at phase 0; F21 and F22 start together	A; controlled worst convergence

Source: the authors.

The factor k applies only to scalable flows F01–F07, F11–F13, and F18–F20 by replacing the nominal period or cycle T with $T' = T/k$. Brake flows F08–F10, window flows F15–F17, and multicast video F14 remain fixed. F21–F22 are modeled as separate transient events and are not multiplied by k.

Under A0, the four camera phases are 0, C/4, C/2, and 3C/4, and the two transient events do not overlap. Under A1, all cameras use phase 0 and F21–F22 start simultaneously. Average loads, packet sizes, routes, and numbers of messages remain unchanged between A0 and A1; only release-time aggregation changes.

F14 is not scaled because its 61.680 Mbit/s stream must remain below the 100 Mbit/s display access capacity. L3 was selected to bring the CSW-to-HPC direction of LC to approximately 86.3% average utilization while remaining physically feasible at the link level and allowing transient overload from aligned bursts. It deliberately crosses the 75% CBS reservation cap at that egress and is therefore a reservation-feasibility boundary for C3 and C4, not an admissible high-load operating point for those configurations.

3.5 PER LINK LOAD ANALYSIS

Per-link loads are calculated independently for each direction because every Ethernet link is full-duplex. Table 5a reports the permanent mean wire load in Mbit/s, separating the fixed component from the scalable component at k = 1 and showing the resulting L0–L3 values. Table 5b preserves the corresponding transient peak increments.

Table 5a – Permanent mean wire load by link and direction (Mbit/s).

Link / direction	Fixed	Scalable k=1	L0	L1	L2	L3
LC: CSW→HPC	0	215.664	107.832	215.664	539.160	862.656
LC: HPC→CSW	91.920	93.120	138.480	185.040	324.720	464.400
LB1: Z1→CSW	0	146.100	73.050	146.100	365.250	584.400
LB1: CSW→Z1	71.760	31.040	87.280	102.800	149.360	195.920
LB2: Z2→CSW	0	23.412	11.706	23.412	58.530	93.648
LB2: CSW→Z2	71.760	31.040	87.280	102.800	149.360	195.920
LB3: Z3→CSW	0	46.152	23.076	46.152	115.380	184.608
LB3: CSW→Z3	10.080	31.040	25.600	41.120	87.680	134.240
LP1: Z1_CPU→Z1_SW	0	0	0	0	0	0
LP2: Z2_CPU→Z2_SW	0	0	0	0	0	0
LP3: Z3_SW→Z3_CPU	0	0	0	0	0	0

Source: the authors.

Table 5b – Transient peak increments by link and direction.

Link / direction	Transient peak increment
LC: CSW→HPC	0; aligned camera arrivals up to 3 Gbit/s
LC: HPC→CSW	0
LB1: Z1→CSW	+111.99 Mbit/s (F21)
LB1: CSW→Z1	0
LB2: Z2→CSW	+111.99 Mbit/s (F22)
LB2: CSW→Z2	0
LB3: Z3→CSW	0
LB3: CSW→Z3	A0 +111.99; A1 +223.97 Mbit/s
LP1: Z1_CPU→Z1_SW	+111.99 Mbit/s (F21)

Link / direction	Transient peak increment
LP2: Z2_CPU→Z2_SW	+111.99 Mbit/s (F22)
LP3: Z3_SW→Z3_CPU	A0 +111.99; A1 +223.97 Mbit/s

Source: the authors.

LC in the CSW-to-HPC direction is the principal mean-load convergence point: it reaches 862.656 Mbit/s at L3. This entire permanent load is carried by CBS-shaped TC6 or TC4 flows F01–F07 and already exceeds the 750 Mbit/s per-port cap before any calibration multiplier is applied. Consequently, C3 and C4 at L3 are preclassified as reservation-infeasible at this egress under the declared rule. L3 remains in the factor matrix to expose this boundary; the classification is not a simulation failure. F14 belongs to the fixed HPC-to-CSW, CSW-to-Z1, and CSW-to-Z2 components and remains at 61.680 Mbit/s across L0–L3.

Mean-load feasibility does not eliminate short overload intervals. Under A1, the four cameras release 30 frames simultaneously, and the aggregate arrival rate at the CSW-to-HPC egress can approach the combined capacity of three zonal uplinks, up to 3 Gbit/s before serialization onto the 1 Gbit/s LC link. F21–F22 add up to approximately 223.97 Mbit/s on the converging Z3 path when aligned. These effects are intentionally represented through queue occupancy, delay, and loss rather than hidden in average utilization.

3.6 TRAFFIC MODEL DECISIONS

The benchmark uses the burst-oriented camera model from Migge et al. (2018): 30 application payloads of 1,446 bytes per image every 16 ms at nominal load. This representation is retained because burst aggregation and source pre-shaping are experimental factors. The alternative periodic camera model is not part of this project.

The Luo et al. (2022) video flow is adapted to multicast from the HPC to two displays, with replication at the central switch. Best-effort traffic uses three controlled unidirectional UDP flows, and the measured duration–volume pairs from Leal Filho et al. (2025) define F21–F22; their endpoints and convergence toward Z3_CPU are benchmark adaptations.

Common closure decisions are: 100 ns propagation per link; store-and-forward switching; physical-rate calculations including Ethernet overhead, preamble/SFD, and IFG; VLAN identifiers 18–22 for the added flows; 20 Mbit/s nominal best-effort load per flow; and

drop-tail queues of 128 KiB per class at the identified bottlenecks, 64 KiB per class at the F02–F05 camera talker egress ports, and 32 KiB per class elsewhere. These choices remain fixed across C0–C4.

4 TSN CONFIGURATION DESIGN

4.1 COMMON CLASSIFICATION AND QUEUEING RULES

C0–C4 use the same topology, routes, workload, deadlines, buffers, and random realizations. Every application frame retains the VLAN identifier and PCP in the Traffic Catalog, and INET maps PCP n directly to traffic class TCn. The benchmark uses TC7 for brake control (F08–F10), TC6 for the 100 µs LiDAR and radar flows (F01, F06–F07), TC4 for cameras, audio, multicast video, and transient ADAS transfers (F02–F05, F11–F14, F21–F22), TC3 for window control, and TC0 for best-effort traffic; TC1, TC2, and TC5 are unused by application traffic. This benchmark-specific grouping separates the 100 µs sensors from the longer-deadline traffic without changing application deadlines or criticality. In C4, gPTP frames are identified by protocol type before the normal PCP mapping and assigned to TC7 without changing the VLAN or PCP values of F01–F22.

Queues are FIFO within each class and, when more than one class is eligible, the higher traffic class is selected first. All treatments use non-preemptive, store-and-forward Ethernet and the same drop-tail capacities. CBS is enabled only for TC6 and TC4 in C2–C4; TAS is enabled only for TC7 on the seven selected egresses along the three brake paths in C4.

4.2 STRICT PRIORITY CONFIGURATIONS

C0 is the Strict Priority baseline and applies no shaping. C1 differs only by spacing the 30 packets of each F02–F05 camera image across its cycle. Packet count, size, average rate, route, PCP, VLAN, deadline, and camera egress queue capacity (64 KiB per class) remain unchanged, so the C0–C1 comparison isolates the effect of reducing camera microbursts.

The spacing is deterministic and limited so that one image burst does not extend beyond the current camera cycle. All non-camera flows retain the release pattern defined in Section 3.

4.3 CREDIT-BASED SHAPING CONFIGURATIONS

C2 and C3 apply the Credit-Based Shaper to two egress queues: TC6 carries F01, F06, and F07; TC4 carries F02–F05, F11–F14, and F21–F22. These are custom CBS traffic classes defined by the benchmark, rather than a reproduction of standard AVB SR-A/SR-B service contracts. TC7 brake, TC3 window, and TC0 best-effort queues remain under Strict Priority. CBS is configured at talker and switch egresses where the respective class has a positive permanent nominal reservation; it is not applied to every traffic class.

For each egress port and shaped class, the nominal reservation is the sum of the permanent wire rates routed through that port. F21–F22 use TC4 while active but are excluded from the permanent reservation because they are transient perturbations. They share CBS service at egresses where TC4 has a positive permanent reservation. Where TC4 carries only transient flows and its permanent nominal reservation is zero, it remains under Strict Priority, preserving PCP, traffic class, FIFO order, and relative priority. No CBS instance with zero idleSlope or additional transient reservation is introduced. This placement rule is common to C2 and C3 and is retained in C4.

C2 uses this nominal per-port reservation. C3 uses the same queue mapping and placement but replaces the nominal values with calibrated reservations. No other workload or queueing rule changes between C2 and C3.

4.4 CBS CALIBRATION

Calibration is performed once at L2/A1. It starts from the nominal reservation and increases the affected egress/class values in deterministic steps until the viability criteria in Section 5.4 are met, subject to a total CBS reservation limit of 75% of the port rate. The values are then reduced in smaller steps to avoid unnecessary reservation.

The resulting calibrated-to-nominal ratios are frozen and applied to the other load levels. Before execution, each C3 cell is screened by applying those ratios to its per-port nominal loads. A cell whose required total CBS reservation exceeds 75% at any egress is recorded as reservation-infeasible and is not retuned, clipped, or interpreted as a simulation failure. This rule preclassifies C3/L3 as infeasible at the CSW-to-HPC egress. Calibration remains a project procedure inspired by Migge et al. (2018), not an analytical optimum or a reproduction of RTaW-Pegase.

4.5 TIME-AWARE SHAPING WITH CREDIT-BASED SHAPING

C4 retains the calibrated CBS queues from C3 and adds Time-Aware Shaping for brake flows F08–F10 at seven egress interfaces: the HPC egress toward the CSW, the three CSW egresses toward ZSW_i, and each ZSW_i egress toward BRAKE_i. A synchronized 100 µs cycle opens TC7 for 10 µs, opens TC0–TC6 for 77 µs, and closes all gates for a 13 µs guard band. The same cycle and gate vector are used at all seven egresses. The guard band exceeds the serialization time of the largest benchmark frame and frame preemption is not used.

The HPC, CSW, and Z1_SW–Z3_SW use the gPTP domain defined in Section 2.5. At the HPC-to-CSW gated egress, the TC7 window carries both brake and gPTP control frames. At each CSW-to-ZSW_i egress, it carries the corresponding brake flow and the gPTP messages used to synchronize the downstream zonal switch; at each ZSW_i-to-BRAKE_i egress, it carries the corresponding brake flow. The BRAKE_i end stations do not execute TAS. CBS remains active for TC6 and TC4 only while the corresponding TAS gate is open. INET 4.6 changes CBS credit only during that open interval (INET FRAMEWORK, [s. d.]a); therefore, reusing the frozen C3 idleSlope values is a controlled baseline, not a guarantee of equivalent service. Before execution, each C4 cell is screened at every gated egress: permanent TC0–TC6 load must fit the 77% open-time budget, and the effective service available to TC6 and TC4 must cover their permanent offered loads. Pilot runs must additionally show stable queues and delivered rates. Failure classifies the cell as gate-capacity-infeasible; CBS and GCL parameters are not retuned.

4.6 CONFIGURATION COMPARISON

Table 6 summarizes the controlled distinction introduced by each treatment. None changes the topology, traffic catalog, load factors, aggregation pattern, deadlines, or seed realization.

Table 6 – Controlled comparison of the C0–C4 configurations.

Configuration	Mechanism and treated traffic	Frozen parameters	Controlled distinction
C0	Strict Priority for PCP/TC 7, 6, 4, 3, and 0; no shaping.	Drop-tail buffers; descending TC/PCP order.	Baseline.
C1	C0 plus source pre-shaping for F02–F05.	$\delta_{ADAS} = \min(324\ \mu s, C/30)$; $\delta_{vision} = \min(500\ \mu s, C/30)$.	Camera release pattern only.

Configuration	Mechanism and treated traffic	Frozen parameters	Controlled distinction
C2	Nominal CBS: PCP/TC6 for F01, F06, and F07; PCP/TC4 for F02–F05, F11–F14, and F21–F22; remaining traffic under SP.	idleSlope = R(p,c,L) where R > 0; SP where R = 0; transients excluded from base reservation; original camera bursts retained.	Adds nominal per-port and per-class reservation.
C3	Same CBS mapping, placement, and source releases as C2.	Calibrated at L2/A1; total CBS reservation limited to 75% of the port rate; calibrated ratios frozen.	Replaces nominal with calibrated reservation.
C4	C3 CBS plus TAS for F08–F10 at seven egresses: HPC-to-CSW, CSW-to-ZSW_i, and ZSW_i-to-BRAKE_i; one gPTP domain for HPC, CSW, and ZSW_i.	100 µs GCL at seven egresses: 10 µs TC7/PCP 7, 77 µs TC0–TC6/PCP 0–6, 13 µs closed; frozen C3 vector used as a candidate baseline; gate-aware capacity check required; gPTP intervals 125 ms and 1 s.	Adds end-to-end synchronized gating, protocol overhead, and a 77% open-time capacity constraint; no CBS or GCL retuning.

Source: the authors.

The comparison therefore changes one declared traffic treatment at a time while preserving the common benchmark.

5 EXPERIMENTAL DESIGN

5.1 EXPERIMENTAL FACTORS AND MATRIX

The experiment crosses three factors: five network configurations (C0–C4), four offered-load levels (L0–L3, with multipliers 0.5, 1.0, 2.5, and 4.0), and two aggregation patterns (A0 staggered and A1 aligned). This produces 40 experimental cells. Load scaling changes the period or camera cycle of the scalable flows defined in Section 3; A0 and A1 change only release alignment, not packet sizes, routes, or average offered rates.

The full design contains 40 logical cells. Four cells—C3 and C4 at L3 under A0 and A1—are retained as reservation-frontier results but are not simulated, because the CSW-to-HPC egress requires 862.656 Mbit/s of permanent CBS reservation before calibration, above the 750 Mbit/s cap. The remaining 36 cells are candidates for 30 replications each, giving at most 1,080 final runs; any additional C4 cell that fails the gate-aware capacity screen is likewise recorded as infeasible without final replications. Topology, routes, link capacities, Traffic Catalog entries, queue capacities, deadlines, and metric definitions remain fixed.

5.2 RUN PROTOCOL

Each final run uses 1 s of warm-up followed by 30 s of measurement. Final seeds are 1001–1030 and are paired across the configurations that are executed for the same load and aggregation condition. The sampled F21–F22 transient sequence is also reused across those configurations for the corresponding seed.

A run is identified by configuration, load, aggregation, and seed. Abnormal termination, missing signals, or invalid configuration marks the run as invalid and requires a documented rerun with the same seed; observations are not removed solely because they are unfavorable.

5.3 CALIBRATION, PILOT, AND FINAL CAMPAIGN

CBS calibration and the integrated pilot use seeds that are separate from the final campaign. Calibration is performed at L2/A1, and the resulting reservation ratios are frozen before final execution. Pre-run screening checks the 75% per-port CBS cap for C3 and C4 and, for C4, the 77% gate-open budget and effective TC6/TC4 service at every gated egress. The pilot then checks traffic generation, classification, shaping, gating, synchronization, metric collection, queue stability, delivered rates, and the adequacy of the selected run duration. For C4, it also confirms that the HPC–CSW and HPC–ZSW_i clock offsets stabilize during the 1 s warm-up, that gPTP frames cross LC and LB without persistent queue buildup or loss, and that F08–F10 traverse all three gated egress stages within the intended TC7 windows.

After pilot acceptance, the topology, Traffic Catalog, C0–C4 parameters, metrics, and analysis rules are frozen. Final results do not trigger CBS retuning, GCL changes, workload adjustment, or selective repetition. A cell rejected by reservation or gate-capacity screening remains an explicit boundary result rather than being forced into an admissible configuration.

5.4 METRICS AND VIABILITY CRITERIA

Primary metrics are end-to-end latency (p99 and maximum observed), deadline miss ratio, useful throughput, packet loss, link utilization, queue occupancy, queueing time, and drops. CBS credit and TAS gate state are collected when needed to explain behavior. In C4, total link utilization includes application and gPTP traffic wherever the protocol is active, whereas useful throughput, latency, and deadline metrics cover only F01–F22; gPTP frames

and bytes on LC and each LB link, together with the maximum absolute HPC–CSW and HPC–ZSW_i clock offsets, are recorded separately. Results are reported per flow and summarized by traffic group, bottleneck, and experimental cell.

A design cell is configuration-admissible only when the declared per-port reservation constraints and, for C4, the gate-aware capacity checks are satisfied. Among executed cells, viability additionally requires that no hard-critical flow miss a deadline, each soft-deadline flow have at most 1% misses, every best-effort flow achieve a received-to-offered application-payload ratio of at least 0.95 with at most 1% packet loss, and no queue grow continuously after warm-up. Maximum observed latency is an empirical result, not an analytical worst-case guarantee.

Configuration effort is discussed using the number of shaped classes and egress instances, the need for CBS calibration, GCL entries, synchronization, and reserved capacity. These indicators are secondary to timing and service results but are needed to interpret practical trade-offs.

5.5 HYPOTHESIS TESTING AND DATA ANALYSIS

Table 7 defines the contrasts used for H1–H5. The run is the unit of replication. Comparisons use paired results for equal seeds, with absolute and relative effects and 95% confidence intervals where applicable.

Table 7 – Predefined hypothesis-to-evidence mapping.

Hypothesis	Predefined contrast	Primary evidence	Interpretation rule
H1	C0–C4 at L0/A0	Critical-flow p99, maximum, and deadline miss ratio	Small differences under low contention support H1.
H2	A1−A0 in C0; C1−C0 at L2/L3	Tail latency, misses, peak queue, and queueing time	Alignment degrades C0 and source pre-shaping reduces the effect.
H3	C3−C2 at L2; L3 reservation boundary	Viability frontier, SR misses, best-effort goodput, and reserved bandwidth	Calibrated CBS sustains viability further without unacceptable residual-service loss.
H4	C4−C3 in capacity-admissible cells	Brake timing; unscheduled wait, effective CBS service, idle gate capacity, and configuration cost	TAS improves brake timing and its residual-service and configuration costs are quantified.
H5	C0–C4 across all logical L/A cells, including infeasibility	Pareto set for timing, best effort, queues, reservation, and complexity	No single configuration dominates every operating regime.

Source: the authors.

Tail latency, deadline misses, and viability are primary; averages are descriptive. The analysis examines how the treatment effect changes with load and burst alignment instead of seeking one global ranking.

A hypothesis is supported only when its predefined contrast has the expected direction and a practically relevant magnitude relative to deadlines, service thresholds, queue behavior, or configuration cost. Quantitative paired contrasts use only cells in which both configurations are admissible and executed. H5 retains reservation- and gate-capacity-infeasible cells as explicit boundary outcomes rather than treating them as missing data when assessing whether any configuration dominates across all operating regimes.

6 IMPLEMENTATION AND VALIDATION PLAN

6.1 IMPLEMENTATION SCOPE

P2 will implement one parameterized OMNeT++/INET model containing the three-zone topology and the 22 catalog flows. C0–C4, L0–L3, A0–A1, seed, warm-up, and measurement duration will be selected through run configurations so that the topology and workload do not need to be edited between scenarios.

Implementation may proceed incrementally, beginning with a small topology that verifies packet generation, forwarding, PCP-to-traffic-class mapping, latency measurement, and queue observation. It is then expanded to the full network, followed by CBS and TAS. The internal source-code and directory organization is not prescribed by P1 as long as the defined scenarios remain selectable and reproducible.

6.2 VERIFICATION AND PILOT

Verification will compare configured and observed application and gPTP message rates, link loads, routes, destinations, traffic classes, queue selection, CBS rate limiting, TAS gate operation at all seven selected egresses, the HPC–CSW and HPC–ZSW_i clock offsets, and collected metrics. Focused test scenarios may be used when the full topology does not make a mechanism easy to observe. For C0 and C1, verification must confirm zero camera-source droppedPacketsQueueOverflow counters, the expected generated frame counts, and per-flow accounting of reception and any network-side drops. After the camera queue correction, both configurations must be revalidated and compared over all eight L0–L3

× A0–A1 combinations, using identical seeds and run settings. Runs produced with 32 KiB camera queues are retained only as debugging evidence and excluded from the final comparison; corrected runs replace them.

Before the final campaign, an integrated pilot will execute all five configurations at L2/A1 with seeds reserved for piloting. The pilot is accepted when every flow follows its catalog definition, every required metric is produced, the intended mechanisms are active, and no unresolved error compromises interpretation. Corrections are allowed after the pilot, but the final configuration and analysis rules are then frozen.

6.3 REPRODUCIBILITY AND HANDOFF

Each run will record the model and dependency versions, configuration, load level, aggregation pattern, seed, durations, and resolved CBS, per-egress TAS, and per-node gPTP parameters. Raw results will be kept separately from processed results, and analysis scripts will produce the tables and figures used in P3 and P4. This information is sufficient to reproduce a scenario without turning P1 into a detailed software specification.

7 CONCLUSION

This design defines a controlled comparison of five traffic treatments in a three-zone automotive Ethernet network. The architecture, 22-flow catalog, traffic-class mapping, load and aggregation factors, metrics, and viability criteria provide the common basis for implementation and evaluation.

P2 will implement the model and experiment runner; P3 will verify the traffic and mechanisms, execute the pilot and final campaign, and analyze the predefined contrasts; P4 will report the results and limitations. The document intentionally fixes the experimental decisions that affect comparability while leaving ordinary software-organization choices to the implementation stage.

REFERENCES

COLUSSI, Leonardo. **Co-design of in-vehicle networks**: a multi-method approach for optimization, simulation, and predictive analysis. 2025. Trabalho de Conclusão de Curso (Graduação em Engenharia de Controle e Automação) – Universidade Federal de Santa Catarina, Florianópolis, 2025.

IEEE. **IEEE Std 802.1Qav-2009**: IEEE Standard for Local and Metropolitan Area Networks—Virtual Bridged Local Area Networks—Amendment 12: Forwarding and Queuing Enhancements for Time-Sensitive Streams. New York: IEEE, 2009. Available at: IEEE 802.1Qav-2009. Accessed on: 24 Sept. 2026.

IEEE. **IEEE Std 802.1Qbv-2015:** IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks—Amendment 25: Enhancements for Scheduled Traffic. New York: IEEE, 2015. Available at: IEEE 802.1Qbv-2015. Accessed on: 24 Sept. 2026.

IEEE. **IEEE Std 802.1AS-2020**: IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications. New York: IEEE, 2020. Available at: IEEE 802.1AS-2020. Accessed on: 24 Sept. 2026.

IEEE. **IEEE Std 802.1Q-2022**: IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks. New York: IEEE, 2022. Available at: IEEE 802.1Q-2022. Accessed on: 24 Sept. 2026.

IEEE. **IEEE Std 802.1DG-2025**: IEEE Standard for Local and Metropolitan Area Networks—Time-Sensitive Networking Profile for Automotive In-Vehicle Ethernet Communications. New York: IEEE, 2025. Available at: IEEE 802.1DG-2025. Accessed on: 24 Sept. 2026.

INET FRAMEWORK. **Combining time-aware and credit-based shaping**. INET Framework Showcase, version 4.6. [S. l.], [s. d.]a. Available at: INET CBS+TAS showcase. Accessed on: 23 Sept. 2026.

INET FRAMEWORK. **Credit-based shaping**. INET Framework Showcase, version 4.6. [S. l.], [s. d.]b. Available at: INET CBS showcase. Accessed on: 23 Sept. 2026.

LEAL FILHO, Josafat; FRÖHLICH, Antônio Augusto; HORSTMANN, Leonardo Passig; HOFFMANN, José Luis Conradi; XAVIER, Jozimar Custódio. Predicting transient overloads related to ADAS in time-sensitive vehicular networks. In: IEEE SYMPOSIUM ON COMPUTERS AND COMMUNICATIONS, 2025. **Proceedings [...]**. [S. l.]: IEEE, 2025. DOI: 10.1109/ISCC65549.2025.11326440.

LUO, Feng; WANG, Bowen; YANG, Zhenyu; ZHANG, Ping; MA, Yifei; FANG, Zihao; WU, Mingzhi; SUN, Zhipeng. Design methodology of automotive time-sensitive network system based on OMNeT++ simulation system. **Sensors**, Basel, v. 22, n. 12, art. 4580, 2022. DOI: 10.3390/s22124580.

MIGGE, Jörn; VILLANUEVA, Josetxo; NAVET, Nicolas; BOYER, Marc. Insights on the performance and configuration of AVB and TSN in automotive Ethernet networks. In: EUROPEAN CONGRESS ON EMBEDDED REAL TIME SOFTWARE AND SYSTEMS (ERTS), 9., 2018, Toulouse. **Proceedings [...]**. Toulouse: SEE, 2018. p. 1–14.

NAVET, Nicolas; VILLANUEVA, Josetxo; MIGGE, Jörn; BOYER, Marc. **Insights on the performance and configuration of AVB and TSN in automotive applications**. In: IEEE STANDARDS ASSOCIATION ETHERNET & IP @ AUTOMOTIVE TECHNOLOGY DAY, 2017, San Jose. Presentation. San Jose: IEEE Standards Association, 2017.